

Nmap Cheat Sheet

Overview: What is Nmap? Why is Nmap useful?

Nmap is an essential open-source tool for Ethical Hackers and Penetration testers. It was initially created by Gordon Lyon (aka Fyodor).

Nmap ("Network Mapper") is a free and open source utility for network discovery and security auditing. Many systems and network administrators also find it useful for tasks such as network inventory, managing service upgrade schedules, and monitoring host or service uptime. Nmap uses raw IP packets in novel ways to determine what hosts are available on the network, what services (application name and version) those hosts are offering, what operating systems (and OS versions) they are running, what type of packet filters/firewalls are in use, and dozens of other characteristics.

Nmap Help

Use `nmap -h` to display an extended help menu of Nmap. This menu provides an overview of all possible options and their required arguments. You can also use `man nmap` for an in-depth manual.

Nmap Targeting

Nmap can scan single targets or multiple targets in various ways:

Command	Description
<code>nmap 192.168.1.1</code>	◆ Scanning a single IP
<code>nmap www.domain.com</code>	🌐 Scanning a hostname
<code>nmap 192.168.1.1-100</code>	📊 Scanning an IP range
<code>nmap 192.168.1.1/24</code>	◆ Scanning a subnet
<code>nmap -iL list.txt</code>	📄 Scanning from a predefined list

Nmap Scan Types

Each scan type has unique capabilities with different noise levels and requirements:

Command	Description	Root/Sudo Required	Noise Level
<code>nmap -sS <target></code>	 TCP SYN SCAN (Stealth scan) - Sends only SYN packets and awaits SYN/ACK responses. Fast and accurate scan used most often.	✅ Required	🟢 Very Low
<code>nmap -sT <target></code>	 TCP Connect Scan - More accurate than SYN scan but slower and noisier. Makes full three-way handshake with the host.	❌ Not Required	🟡 Medium
<code>nmap -sU <target></code>	 UDP Scan - Scans for UDP ports. Slower and more difficult but useful for services like DNS, SNMP, DHCP.	✅ Required	🟡 Medium
<code>nmap -sn <target(s)></code>	 Ping Scan - Simple ping scan to see which hosts reply to ICMP packets. Fast way to discover live hosts.	❌ Not Required	🟢 Very Low
<code>nmap -sV <target></code>	 Service Version Scan - Detects service versions behind open ports. Helpful for exploit selection.	❌ Not Required	🟡 Medium
<code>nmap -O <target></code>	 OS Detection - Detects remote operating system. Requires at least 1 open and 1 closed port.	✅ Required	🟡 Medium
<code>nmap -A <target></code>	 Aggressive Scan - Combines OS detection, version detection, script scanning, and traceroute. Very noisy!	✅ Required	🔴 Very High

Nmap Port Scanning

Specify which ports to scan with these options:

Command	Description
<code>nmap -p <port> <target></code>	 Scan for one specific port
<code>nmap -p <start>-<end> <target></code>	 Scan a range of ports (e.g., <code>-p 1-20</code>)
<code>nmap -p <port_a>,<port_b>,<port_c> <target></code>	 Scan multiple specific ports (comma-separated)
<code>nmap -p U:<udp_port>,T:<tcp_port> <target></code>	 Scan both UDP and TCP ports
<code>nmap -F <target></code>	 Fast scan - 100 most common ports
<code>nmap --top-ports <amount> <target></code>	 Scan top # ports (specify amount)

Command	Description
<code>nmap -p- <target></code>	🌐 Scan all 65,535 ports (very slow!)

🕒 Nmap Timing Options

Control scan speed and stealth with timing templates (T0-T5):

Command	Description
<code>nmap -T0 <target></code>	🦹 Paranoid - Slowest scan, best for IDS evasion
<code>nmap -T1 <target></code>	😬 Sneaky - Slow scan, good for IDS evasion
<code>nmap -T2 <target></code>	👮 Polite - Reduces bandwidth usage and target resources
<code>nmap -T3 <target></code>	⚖ Normal - Default timing (same as no template)
<code>nmap -T4 <target></code>	🏃 Aggressive - Faster scan for reliable networks
<code>nmap -T5 <target></code>	🚀 Insane - Fastest mode, may sacrifice accuracy

📄 Nmap Scripts

Nmap provides powerful NSE (Nmap Scripting Engine) scripts organized by categories:

Script Categories:

- 🔑 **auth** - Authentication scripts
- 📡 **broadcast** - Network broadcast scripts
- ★ **default** - Default safe scripts
- 🔍 **discovery** - Host and service discovery
- 💣 **dos** - Denial of service detection
- 🎯 **exploit** - Exploitation scripts
- 🌐 **external** - External service scripts
- 🛠 **fuzzer** - Fuzzing scripts
- ⚠ **intrusive** - Intrusive scripts
- 🦠 **malware** - Malware detection
- ✅ **safe** - Safe scripts
- 📋 **version** - Version detection
- 🛡 **vuln** - Vulnerability detection

Usage:

```
nmap --script <script/script_group> <target>
```

⚠ **Note:** Script scans require root/sudo privileges and vary in noise levels. Always read script descriptions before use.

🎁 Extras & Additional Resources

📄 Output to File

Save scan results to a text file for reports:

```
nmap -oN <filename.txt> <target>
```

🔥 Common Combined Scan

Most frequently used scan combination:

```
sudo nmap -sS <target> -oN <filename.txt>
```

📷 Command Examples

Basic Scan:

```
nmap 192.168.0.239
```

```

[REDACTED]@ [REDACTED]:~/Desktop$ nmap 192.168.0.239
Starting Nmap 7.80 ( https://nmap.org ) at 2020-11-28 05:59 CST
Nmap scan report for 192.168.0.239
Host is up (0.00034s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 11.11 seconds
[REDACTED]@ [REDACTED]:~/Desktop$

```

Specific Port Scan:

```
nmap -p 80 192.168.0.239
```

```

[REDACTED]@ [REDACTED]:~/Desktop$ nmap -p 80 192.168.0.239
Starting Nmap 7.80 ( https://nmap.org ) at 2020-11-28 06:10 CST
Nmap scan report for 192.168.0.239
Host is up (0.00047s latency).

PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 11.06 seconds
[REDACTED]@ [REDACTED]:~/Desktop$

```

All Ports Scan:

```
nmap -p- 192.168.0.239
```

```
@:~/Desktop$ nmap -p- 192.168.0.239
Starting Nmap 7.80 ( https://nmap.org ) at 2020-11-28 06:11 CST
Nmap scan report for 192.168.0.239
Host is up (0.00020s latency).
Not shown: 65505 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
33179/tcp open  unknown
44399/tcp open  unknown
45805/tcp open  unknown
51579/tcp open  unknown

Nmap done: 1 IP address (1 host up) scanned in 12.51 seconds
@:~/Desktop$
```

Service Version Scan:

```
sudo nmap -sV 192.168.0.239
```

```

@:~/Desktop$ sudo nmap -sV 192.168.0.239
Starting Nmap 7.80 ( https://nmap.org ) at 2020-11-28 06:05 CST
Nmap scan report for 192.168.0.239
Host is up (0.00014s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:9F:F3:C9 (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.62 seconds
@:~/Desktop$

```

Fast Aggressive Scan:

```
nmap -T4 192.168.0.239
```

```

@:~/Desktop$ nmap -T4 192.168.0.239
Starting Nmap 7.80 ( https://nmap.org ) at 2020-11-28 06:19 CST
Nmap scan report for 192.168.0.239
Host is up (0.0016s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 11.11 seconds
@:~/Desktop$

```

Stealth SYN Scan:

```
sudo nmap -sS 192.168.0.239
```

```
██████████@██████████:~/Desktop$ sudo nmap -sS 192.168.0.239
Starting Nmap 7.80 ( https://nmap.org ) at 2020-11-28 06:02 CST
Nmap scan report for 192.168.0.239
Host is up (0.000081s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:9F:F3:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 11.39 seconds
██████████@██████████:~/Desktop$
```

Ping Scan:

```
nmap -sn 192.168.0.239
```

```
██████████@██████████:~/Desktop$ nmap -sn 192.168.0.239
Starting Nmap 7.80 ( https://nmap.org ) at 2020-11-28 06:04 CST
Nmap scan report for 192.168.0.239
Host is up (0.00039s latency).
Nmap done: 1 IP address (1 host up) scanned in 11.03 seconds
██████████@██████████:~/Desktop$
```

Additional Learning Resources

-  Check out Gordon Lyon's *Nmap Network Scanning* book
-  Official Nmap documentation: <https://nmap.org/docs.html>
-  Practice in controlled environments only
-  Always follow ethical hacking guidelines and obtain proper authorization

Remember: Use these tools responsibly and only on systems you own or have explicit permission to test. Ethical hacking requires proper authorization and adherence to legal boundaries.

🛡️ Happy Ethical Hacking! 🛡️